

Statistics

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Outline

Statistics

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Amrutsamanvar

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Population and
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Levels of
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About Me – The Serious (and Not-So-Serious) Version

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- Rushikesh
- Born in the Western Ghats of India.
- IIT Madras alumnus
 - ▶ just lost part of my soul and hair over there
- Worked at TU Dresden and collaborated with MIT, Yale, UC Berkeley and UT Austin
 - ▶ Cool, isn't it?
- Lifelong learner
- Beyond the Nerd Life: I love history and culture, travel, lift weights, ride bikes, and burn things in the kitchen - sometimes intentionally

"Life is way beyond our professions"

Welcome to the class!

Agenda

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- What to expect?
- Interactive classes
- Hands-on
- Handling real-world problems
- Mini group discussions
- Enjoying the learning process

Let's Think...

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- If I say there is a **70% chance of rain**, what does it really mean?
- If a medical test is **99% accurate**, can it still be wrong most of the time?
- Who believes they are **above average** in intelligence or driving?
- Can we **mislead people using true data**?

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Why learn it?

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Statistics is the science of **learning from data**, making decisions, and understanding uncertainty.

- We are surrounded by data — but data alone is not useful
- Decisions in business, science, and policy rely on data
- If you cannot interpret uncertainty → you cannot make informed decisions
- Statistics helps you:
 - ▶ summarize data
 - ▶ identify patterns
 - ▶ quantify uncertainty
 - ▶ test assumptions
 - ▶ avoid misleading conclusions

Insight

“Without statistics, data is just numbers — not insight.”

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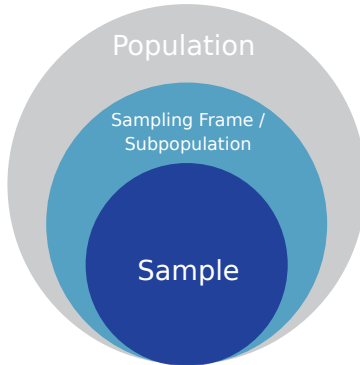
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Population, Sampling Frame, and Sample



Target population which
needs to be studied



Accessible part
of target population



Actual observations

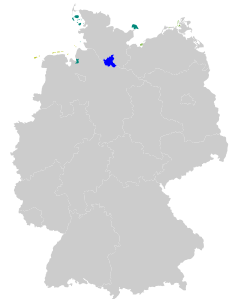
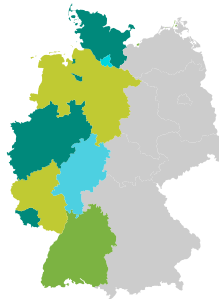
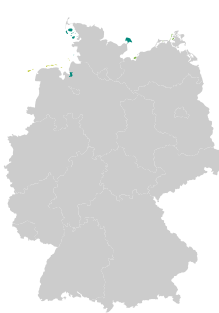
Population, Sampling Frame, and Sample



Target population which
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Accessible part
of target population



Actual observations

Qualitative and Quantitative Data

Understanding data types

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Quantitative Data



- Numerical values
- Can be measured
- Precise and objective

Qualitative Data



- Descriptive information
- Categories and types
- Based on observation

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Nature of Data

- **Quantitative**

- ▶ Numerical (measured/count)
- ▶ *Age, Income, Returns*

- **Qualitative**

- ▶ Categories/types
- ▶ *Gender, Nationality*

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Nature of Data

● Quantitative

- ▶ Numerical (measured/count)
- ▶ *Age, Income, Returns*

● Qualitative

- ▶ Categories/types
- ▶ *Gender, Nationality*

Value Structure

● Continuous

- ▶ Any value (no gaps)
- ▶ *Height, Time*

● Discrete

- ▶ Countable (with gaps)
- ▶ *No. of students, Age (years)*

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Types of Characteristics

Organizing data effectively

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Special Cases

Quasi-continuous

Discrete but treated as continuous

- *Income, Returns*

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Special Cases

Quasi-continuous

Discrete but treated as continuous

- *Income, Returns*

Descriptive

Multiple values per unit

- *Mobile numbers*

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Levels of Measurement

The foundation for choosing statistics

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Nominal	Ordinal	Interval	Ratio
Eye color	Level of satisfaction	Wage	Height
Named	Named	Number	Number
	Natural order	Natural order	Natural order
		Equal intervals	Equal intervals
			True zero

- Nominal: Categories only
- Ordinal: Categories + Order
- Interval: Categories + Order + Equal spacing
- Ratio: + True zero (all math possible)

Interval vs Ratio Scales

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Interval Scale

- Difference between values is meaningful
- No true zero point
- Examples:
 - ▶ Temperature ($^{\circ}\text{C}$, $^{\circ}\text{F}$)
 - ▶ IQ Scores
 - ▶ Years (e.g., 2000, 2010)

Ratio Scale

- Difference and ratios are meaningful
- True zero exists
- Examples:
 - ▶ Weight (kg)
 - ▶ Height (cm)
 - ▶ Time duration (minutes)
 - ▶ Income (\$)

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Interval vs Ratio Scale Examples

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Interval Scale

Variable	Explanation
Temperature (°C, °F)	0° is arbitrary, differences matter
IQ Scores	0 IQ $\neq$ absence of intelligence
Calendar Years	0 year is arbitrary
SAT Scores	Differences meaningful, 0 $\neq$ no aptitude
Time of Day	0 AM is arbitrary
pH Levels	Differences matter, 0 $\neq$ no acidity/basicity
Credit Score	0 $\neq$ no creditworthiness

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Ratio Scale

Variable	Explanation
Weight (kg, lbs)	0 = no weight; ratios meaningful
Height (cm, m)	0 = no height; ratios meaningful
Income (\$)	0 = no income; ratios meaningful
Time Duration	0 = no time; ratios meaningful
Distance (km, m)	0 = no distance; ratios meaningful
Age (years)	0 = birth; ratios meaningful
Number of Items	0 = none; ratios meaningful

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Beer Consumption at Oktoberfest

Frequency distributions explained

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Liters Consumed	Frequency (f)	Relative Freq (f/n)	Cumulative
1	2	0.0156	0.0156
2	30	0.2344	0.2500
3	37	0.2891	0.5391
4	28	0.2188	0.7579
5	23	0.1797	0.9376
6	8	0.0625	1.0000

Understanding the Data

- Absolute frequency = number of visitors in each group
- Relative frequency = fraction of total (yellow highlighted)
- Cumulative frequency = running sum up to that row

Key Finding

Most visitors consumed **3 liters** (modal consumption = 37 visitors)

Beer Consumption at Oktoberfest

Visualization of frequency data

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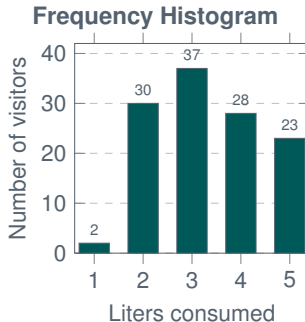
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Summary Statistics:

Liters	Freq	Rel. Freq	Cum.
1	2	0.016	0.016
2	30	0.234	0.250
3	37	0.289	0.539
4	28	0.219	0.758
5	23	0.180	0.938



Interpretation

28.9% of visitors consumed 3 liters.
53.9% consumed up to 3 liters.

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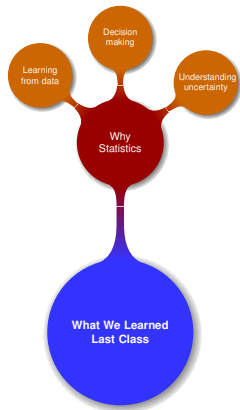
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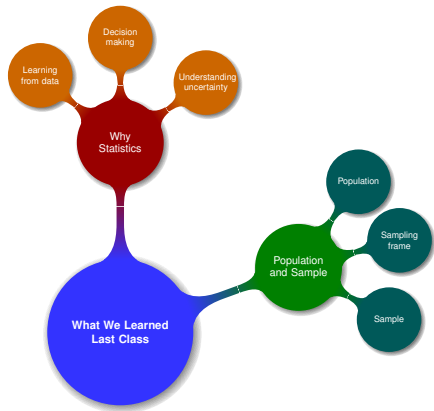
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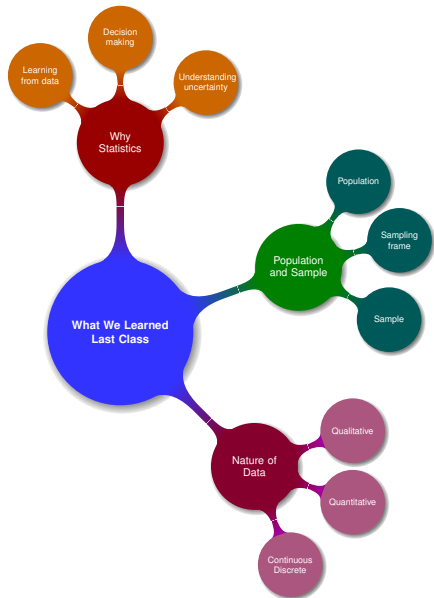
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Our Class Example

20 students: height, gender, and region

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Research Question

If we measure the **height** of 20 students in class, what can frequency, relative frequency, and cumulative frequency tell us?

ID	G	Ht	Reg
1	F	156	EU
2	M	172	SA
3	F	161	EU
4	M	168	ME
5	F	158	SA
6	M	181	AF
7	F	165	EU
8	M	174	EU
9	F	169	SA
10	M	177	ME

ID	G	Ht	Reg
11	F	160	AF
12	M	171	EU
13	F	163	ME
14	M	166	SA
15	F	157	EU
16	M	179	AF
17	F	164	SA
18	M	173	EU
19	F	162	ME
20	M	170	EU

- **G** = Gender (F/M), **Ht** = Height (cm), **Reg** = Region
- Regions: EU, SA, ME, AF (Europe, South America, Middle East, Africa)

How Do We Form Bins?

Step 0: Turn raw heights into groups

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Raw heights from our class

156, 172, 161, 168, 158, 181, 165, 174, 169, 177, 160, 171, 163, 166, 157, 179, 164, 173, 162, 170

- Minimum height = **156 cm**, maximum height = **181 cm**
- Range = $181 - 156 = 25$
- To make the data easier to read, we create equal-width groups of about **5 cm**
- So the bins become:

Bin Width

$$h = \frac{\text{Range}}{k}$$

Here:

$$h \approx \frac{25}{5} = 5$$

155–159	160–164	165–169	170–174	175–179	180–184
---------	---------	---------	---------	---------	---------

Why do this?

- Instead of 20 separate numbers, we now have a few meaningful intervals that are easy to count and easy to plot in a histogram.
- If a student asks about **159.5 cm** then?
 - ▶ anything ≤ 159.5 in lower bin i.e. 159
 - ▶ anything ≥ 159.5 in lower bin i.e. 160

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From Raw Data to Frequency Table

Step 1: Group the heights

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Height Group	Frequency f	Relative Freq.	Cumulative Freq.	Meaning
155–159	3	$3/20 = 0.15$	0.15	15% in this group
160–164	6	$6/20 = 0.30$	0.45	45% are ≤ 164 cm
165–169	4	$4/20 = 0.20$	0.65	65% are ≤ 169 cm
170–174	5	$5/20 = 0.25$	0.90	90% are ≤ 174 cm
175–179	1	$1/20 = 0.05$	0.95	95% are ≤ 179 cm
180–184	1	$1/20 = 0.05$	1.00	100% included

- **Frequency:** how many students fall in each height group.
- **Relative frequency:** the fraction or percentage in that group.
- **Cumulative frequency:** running total from the smallest group upward.
- Great live question: “**What does 0.65 mean here?**”

Choosing the Right Chart

When to use histogram or pie chart

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Histogram

Use for: numerical data grouped into intervals

- Best for **continuous/grouped** data
- Examples: height, weight, income, age
- Shows **distribution** of values
- Helps see concentration and spread

Answers: How are values distributed?

Rule of Thumb

Numerical → Histogram

Categorical → Pie Chart

Pie Chart

Use for: categorical parts of a whole

- Best for **qualitative/categorical** data
- Examples: gender, region, nationality
- Shows **share/proportion** per category
- Helps compare how whole divides

Answers: How is the whole split?

How Do We Plot This?

Pie chart for gender, histogram for height

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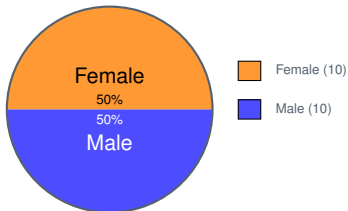
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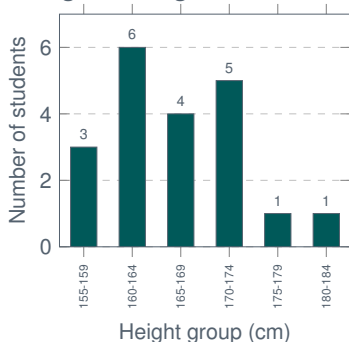
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Pie Chart: Gender Distribution



- Females: 10 (50%)
- Males: 10 (50%)

Histogram: Height Distribution



Observation

Note: Height patterns may differ by region and gender

Adding Region/Nationality

Share of students and average height

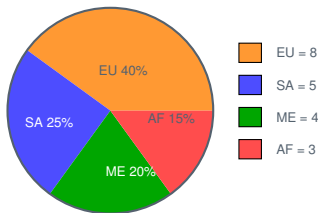
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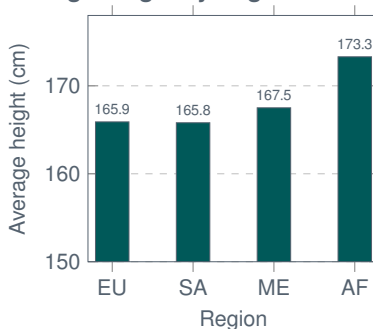
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Regional Distribution



40% from Europe
25% from South America

Average Height by Region



Key Insight

Now we're asking whether patterns change across regions — comparing subgroups!

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Univariate Statistics

Analyzing one variable at a time

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Definition

Univariate statistics means analyzing **one variable** independently.

Variables we can analyze:

- Only height
- Only gender
- Only region

In This Section

- 1 Scale levels & statistics
- 2 Central tendency
- 3 Dispersion

Focus

We'll primarily use **height** to illustrate these concepts.

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2.1 Scale Levels and Statistics

Which statistics fit which scale?

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Main point

Our **height** variable is on a **ratio scale**, so all the common univariate statistics are meaningful.

Scale	Example from class	Useful statistics	What makes sense?
Nominal	Gender, Region	Frequency, Relative Frequency, Mode	Counts and proportions only
Ordinal	Satisfaction level	Median, Percentiles, Rank-based summaries	Order matters
Interval	Temperature in °C	Mean, Variance, Std. Dev.	Differences meaningful
Ratio	Height, Weight, Income	Mean, Median, Range, Variance, Std. Dev.	Zero is meaningful

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2.2 Measures of Central Tendency

Where is the center of the height data?

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Sorted Heights (cm)

156, 157, 158, 160, 161,
162, 163, 164, 165, 166,
168, 169, 170, 171, 172,
173, 174, 177, 179, 181

Calculation Formulas

$$\bar{x} = \frac{\sum x_i}{n}$$

Median = middle value

Results:

M: Mean = **167.3 cm**

$$\frac{3346}{20} = 167.3$$

Md: Median = **167 cm**

$$\frac{166 + 168}{2} = 167$$

Mo: Mode: **160–164 cm**
(grouped modal class)

Interpretation

The typical height in class clusters around **167 cm**

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The Three Pillars of Data Summarization

- 1 **Scale level** → tells which statistics are meaningful
- 2 **Central tendency** → tells the typical value
- 3 **Dispersion** → tells how much variation exists

For our height data (n=20):

- Mean = 167.3 cm
- Median = 167 cm
- Range = 25 cm
- IQR = 12 cm
- Modal class: 160–164 cm

Takeaway

These six numbers summarize the entire distribution of 20 height measurements!

Why do we need summaries?

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The Challenge

Imagine you collect ages of 10 people:

25, 21, 18, 37, 56, 89, 46, 23, 21, 34

Hard to understand quickly!

Our Goal

Find **ONE number** to summarize
Where is the data concentrated?

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Step 1: Sorting Data

First crucial step

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Original Data

25, 21, 18, 37, 56, 89, 46, 23, 21, 34

Sorted Data

18, 21, 21, 23, 25, 34, 37, 46, 56, 89

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Why sorting matters:

- 1 Helps us find the middle value
- 2 Shows which values repeat
- 3 Makes patterns visible

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Mode (Most Frequent Value)

Which value repeats most?

Sorted Data

18, 21, 21, 23, 25, 34, 37, 46, 56, 89

- Which value appears most? 21
- How many times? 2 times
- All others? Appear once

Mode = 21

Meaning

21 is the most common/typical age in this dataset

Median (Middle Value)

The central position

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Sorted Data (n=10, even)

18, 21, 21, 23, 25, 34, 37, 46, 56, 89

- How many values? 10 (even)
- Middle positions? 5th and 6th
- Those values? 25 and 34

$$\text{Median} = \frac{25 + 34}{2} = 29.5$$

Meaning

The median splits the dataset into two equal halves

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Mean (Average)

The sum divided by the count

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Formula

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

For our data:

$$\text{Mean} = \frac{25 + 21 + 18 + 37 + 56 + 89 + 46 + 23 + 21 + 34}{10} = \frac{370}{10} = 37$$

Key Properties

- Uses **all values** in calculation
- Affected by large outliers (like 89)
- Most commonly used - but can be misleading

Key Insight

Comparing the three measures

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Mode	Median	Mean
21 Most common	29.5 Middle	37 Average

Critical Observation:

- Mode (21) < Median (29.5) < Mean (37)
- Mean is **pulled higher** by the extreme value 89
- Median is more **robust** to outliers

Mode (Frequency Table - Step-by-Step)

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- Step 1: Look at frequencies

Value	Frequency
1	2
2	30
3	37
4	28
5	23
6	8

- Highest frequency = 37

Mode = 3

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Median (Frequency Table - Step-by-Step)

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Step 1: Total observations

$$N = 128$$

Step 2: Middle position

$$N/2 = 64$$

Step 3: Build cumulative frequency

Value	Frequency	Cum. Freq
1	2	2
2	30	32
3	37	69
4	28	97

64th observation lies in value = 3
Median = 3

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Mean (Frequency Table - Full Calculation)

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$$\text{Mean} = \frac{\sum (\text{value} \times \text{frequency})}{\sum \text{frequency}}$$

Value	Frequency	Value × Frequency
1	2	1 × 2 = 2
2	30	2 × 30 = 60
3	37	3 × 37 = 111
4	28	4 × 28 = 112
5	23	5 × 23 = 115
6	8	6 × 8 = 48

Total = 448

$$\text{Mean} = \frac{448}{128} = 3.5$$

Mode in Classified Frequency-Table Data

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Frequencies

bin	Frequency
0–1:	3
1–2:	4
2–3:	4
3–4:	2
4–5:	7

- Highest frequency = 7
- Most popular group = 4–5

We don't know exact values!
So we take the middle:

$$\text{Mode} = 4 + 5/2 \approx 4.5$$

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Common Mistake

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- Forgetting the cumulative frequency
- Choosing the wrong median class

Always:

- Find $N/2$
- Use cumulative frequency

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Final Summary

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- Mean = average of all values
- Median = middle value
- Mode = most frequent value

For grouped data:

- We approximate using the middle of the group

Big Idea:

Where is the data concentrated?

What Did We Learn? (Big Picture)

Three types of data distributions

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Data Types We Can Analyze

Raw Data	Individual values (no aggregation)
Grouped Data	Values with frequencies in a table
Classified Data	Ranges/bins with frequencies

How to Calculate Mode, Median, Mean

Data Type	Mode	Median	Mean
Raw	Most frequent value	Middle value	Average of all
Frequency	Highest count value	Use cum. frequency	Weighted mean
Grouped	Midpoint of modal group	Find group, interpolate	Midpoint $\times$ frequency

Memory Aid

Mode = Biggest | **Median** = Middle | **Mean** = Average

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Quantiles

What are they and why do we need them?

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Definition

Quantiles are values that divide a dataset into equal parts.
They show how data is distributed across the range.

Common Quantiles:

- **Percentiles** (100 parts)
- **Quartiles** (4 parts)
- **Deciles** (10 parts)
- **Median** (2 parts)

Why Use Quantiles?

- 1 Understand data spread
- 2 Identify outliers
- 3 Compare distributions
- 4 Make rankings

Key Insight

Quantiles tell us: *What value separates the bottom 25% from the top 75%?*

The Median Revisited

The most important quantile

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What is it?

The **median** is the 50th percentile or second quartile (Q2).
It divides data into two equal halves.

Example with heights (sorted):

156	157	158	160	161	162	163	164	165	166
168	169	170	171	172	173	174	177	179	181

$$\text{Median} = \frac{166 + 168}{2} = 167 \text{ cm}$$

Meaning

Half the class (50%) is below 167 cm, and half is above 167 cm.

Quartiles

Special percentiles that divide data into 4 equal parts

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Definition

Quartiles divide a dataset into four equal groups of 25% each.

The Four Quartiles:

- Q_1 = **First Quartile** = 25th percentile
→ 25% of data is below Q_1 , 75% is above
- Q_2 = **Second Quartile** = 50th percentile = **Median**
→ 50% of data is below Q_2 , 50% is above
- Q_3 = **Third Quartile** = 75th percentile
→ 75% of data is below Q_3 , 25% is above
- Q_4 = **Maximum value**
→ All data is at or below this value

Memory Aid

Quartiles divide into 4 equal parts (Qu-4-ile)

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Quartiles in Detail

Hands-on calculation

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Sorted Heights (n=20):

156, 157, 158, 160, 161, 162, 163, 164, 165, 166, 168, 169, 170, 171, 172, 173,
174, 177, 179, 181

Position calculations:

$$Q_1 \text{ position} = 0.25 \times (n + 1) \\ = 0.25 \times 21 = 5.25$$

$$Q_2 \text{ position} = 0.50 \times 21 = 10.5$$

$$Q_3 \text{ position} = 0.75 \times 21 = 15.75$$

Finding values:

- Q_1 : between 5th and 6th values
= $(161 + 162)/2 = \mathbf{161.5}$ cm
- Q_2 : between 10th and 11th values
= $(166 + 168)/2 = \mathbf{167}$ cm
- Q_3 : between 15th and 16th values
= $(172 + 173)/2 = \mathbf{172.5}$ cm

Interpretation

$Q_1 = 161.5$ | $Q_2 = 167$ | $Q_3 = 172.5$ → Shows the spread of your heights!

Percentiles Across the Distribution

Visual representation

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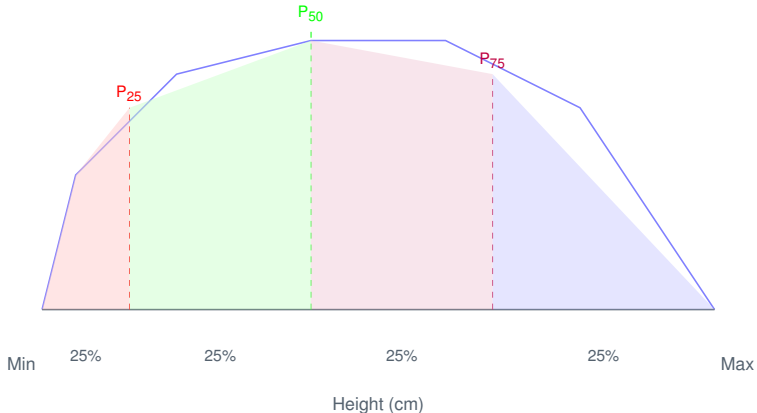
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Each percentile divides the dataset: 25% of data falls in each shaded section between quartiles.

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Interquartile Range (IQR)

The spread of the middle 50%

Definition

The **Interquartile Range (IQR)** is the distance between Q_1 and Q_3 .

$$\text{IQR} = Q_3 - Q_1$$

It contains the **middle 50%** of your data.

For our height data:

$$\text{IQR} = 172.5 - 161.5 = 11 \text{ cm}$$

What it tells us:

- The middle 50% of heights span 11 cm
- Smaller IQR = less spread (more consistent)
- Larger IQR = more spread (more variation)

Comparison

- Range = 25 cm (all data)
- IQR = 11 cm (middle 50%)

Why IQR Matters

IQR ignores extreme outliers — more robust than Range!

Understanding the Five-Number Summary

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The Five Numbers

- 1 **Minimum** = Lowest value
- 2 **Q_1** = 25th percentile
- 3 **Median (Q_2)** = 50th percentile
- 4 **Q_3** = 75th percentile
- 5 **Maximum** = Highest value

For our heights:

Min = 156 | Q_1 = 161.5 | Q_2 = 167 | Q_3 = 172.5 | Max = 181

Power of Five Numbers

These 5 numbers completely describe the distribution and spread of 20 measurements!

Uses:

- ✓ Creating box plots
- ✓ Identifying outliers
- ✓ Comparing distributions
- ✓ Understanding skewness

Box Plot

Visualizing quantiles

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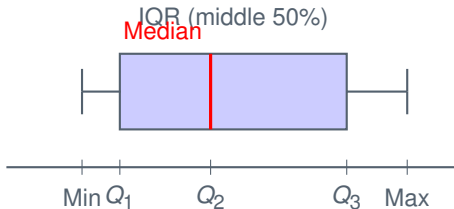
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Box plot shows:

- Box = IQR (where middle 50% of data lies)
- Line in box = Median
- Whiskers = Range from Min to Max

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Percentiles

Dividing data into 100 parts

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Definition

The **pth percentile** is a value below which p% of the data falls.

Common Percentiles:

- P_{25} = 25th percentile (1st quartile) — 25% of data lies below this point
- P_{50} = 50th percentile (Median) — 50% of data lies below this point
- P_{75} = 75th percentile (3rd quartile) — 75% of data lies below this point
- P_{90} = 90th percentile — 90% of data lies below this point (top 10%)

Real-world example — SAT scores:

- If your score is at the **85th percentile**
- → You scored better than 85% of test-takers
- → Only 15% scored higher than you

Key Property

All percentiles together show the **entire distribution** of your data!

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Calculating Percentiles

General formula

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The Percentile Position Formula

$$\text{Position of } P_p = \frac{p}{100} \times (n + 1)$$

where p is the percentile (0 to 100) and n is the sample size.

Step-by-step process:

- 1 Calculate the position
- 2 If position is whole number, take that value
- 3 If position has decimal, interpolate between two values

Example: Find 90th percentile (n=20)

$$\text{Position} = \frac{90}{100} \times 21 = 18.9$$

≈ between 18th and 19th values

$$= 177 + 0.9 \times (179 - 177) = 177 + 1.8 = 178.8 \text{ cm}$$

Meaning

90% of students are 178.8 cm or shorter (top 10% are taller)

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Identifying Outliers with Quantiles

The IQR method

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Outlier Definition (IQR Method)

Lower Bound: $Q_1 - 1.5 \times \text{IQR}$

Upper Bound: $Q_3 + 1.5 \times \text{IQR}$

Any value outside these bounds is an outlier.

For our data:

$$Q_1 = 161.5, \quad Q_3 = 172.5, \quad \text{IQR} = 11$$

$$\text{Lower Bound} = 161.5 - (1.5 \times 11) = 145 \text{ cm}$$

$$\text{Upper Bound} = 172.5 + (1.5 \times 11) = 189.5 \text{ cm}$$

Result

All heights (156–181 cm) fall within $[145, 189.5]$ → No outliers detected!

If we had a height of 220 cm:

- $220 > 189.5$ → It's an outlier (unusually tall)
- Flag for investigation

Practical Example: Comparing Two Distributions

Using quantiles

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Morning class heights vs. Evening class heights

Statistic	Morning	Evening
Min	156	158
Q_1	161.5	163
Median	167	168
Q_3	172.5	173.5
Max	181	185
IQR	11	10.5
Range	25	27

Insights:

- Both groups have similar medians (≈ 167 – 168 cm)
- Evening class has taller maximum (185 vs 181)
- Morning class has slightly larger IQR (less consistency in evening)
- Overall similar distributions

Common Quantile Calculations

Step-by-step guide

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Age Data (Sorted): 18, 21, 21, 23, 25, 34, 37, 46, 56, 89 (n=10)

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Age Data (Sorted): 18, 21, 21, 23, 25, 34, 37, 46, 56, 89 (n=10) **Find Q_1 (25th percentile):**

$$\text{Position} = 0.25 \times (10 + 1) = 2.75$$

$$\begin{aligned}\text{Value} &= 21 + 0.75 \times (21 - 21) \\ &= 21 + 0 = 21 \text{ years}\end{aligned}$$

Find Q_3 (75th percentile):

$$\text{Position} = 0.75 \times 11 = 8.25$$

$$\begin{aligned}\text{Value} &= 46 + 0.25 \times (56 - 46) \\ &= 46 + 2.5 = 48.5 \text{ years}\end{aligned}$$

IQR:

$$\text{IQR} = 48.5 - 21 = 27.5 \text{ years}$$

Key Point

This process works for ANY dataset, no matter the size! 25% of ages are ≤ 21 years, and 75% are ≤ 48.5 years.

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Quantiles vs Other Measures

Side-by-side comparison

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Measure	What it measures	When to use
Median	Middle value (50th percentile)	Always — robust to outliers
Quartiles	4-way split of data	Understand spread, create box plots
Percentiles	Rank in population	Standardized tests, rankings
IQR	Spread of middle 50%	Detect outliers, compare variability
Range	Difference min to max	Quick overview, sensitive to outliers
Std. Dev	Average spread from mean	Statistical tests, normal distribution

Recommendation

Use **five-number summary** (Min, Q_1 , Median, Q_3 , Max) to get a complete picture!

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Summary: Everything About Quantiles

Key Definitions

- **Quantiles** = Values that divide sorted data into equal parts
- **Percentiles** = Quantiles that divide into 100 parts (p%)
- **Quartiles** = Quantiles that divide into 4 parts (Q_1 , Q_2 , Q_3)
- **IQR** = $Q_3 - Q_1$ = Spread of middle 50%

Calculation Steps

- 1 **Sort** the data
- 2 **Find position** using: $\text{Position} = \frac{p}{100} \times (n + 1)$
- 3 **Interpolate** if position has decimal
- 4 **Interpret** in context

Big Picture

Quantiles organize data into ordered groups, revealing the **shape and spread** of your distribution without being fooled by outliers!

Why Special Method for Median in Classified Data?

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- Exact values are not known
- Data is grouped into class intervals
- We estimate using interpolation

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Remember!!

Median is the **middle value** of a dataset.

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Example Table

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Class Interval	Frequency	Cumulative Frequency
0–10	5	
10–20	9	
20–30	14	
30–40	8	
40–50	4	

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- 1 Calculate cumulative frequencies
- 2 Identify the class where the median is located
- 3 Identify the required values from the table to plug into the formula
- 4 Substitute the required entities

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Step 1: Get Cumulative Frequency

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Class Interval	Frequency	Cumulative Frequency
0–10	5	
10–20	9	
20–30	14	
30–40	8	
40–50	4	

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Step 1: Get Cumulative Frequency

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Class Interval	Frequency	Cumulative Frequency
0–10	5	5
10–20	9	14
20–30	14	28
30–40	8	36
40–50	4	40

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Step 2: Identify the class where Median is located

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Class Interval	Frequency	Cumulative Frequency
0–10	5	5
10–20	9	14
20–30	14	28
30–40	8	36
40–50	4	40

- Total frequency: $N = 5 + 9 + 14 + 8 + 4 = 40$
- Median position = $\frac{N}{2} = \frac{40}{2} = 20$
- No look for the row where cumulative frequency became greater than 20
- Median class = **20–30**

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Step 3: Use Median Formula

Identify the entities

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Class Interval	Frequency	Cumulative Frequency
0–10	5	5
10–20	9	14
20–30	14	28
30–40	8	36
40–50	4	40

Median Formula

$$\text{Median} = L + \left(\frac{\frac{N}{2} - CF}{f} \right) \times h$$

- L = lower boundary of median class
- CF = cumulative frequency before median class
- f = frequency of median class
- h = class interval

Step 3: Use Median Formula

Identify the entities

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Class Interval	Frequency	Cumulative Frequency
0-10	5	5
10-20	9	14
20-30	14	28
30-40	8	36
40-50	4	40

Median Formula

$$\text{Median} = L + \left(\frac{\frac{N}{2} - CF}{f} \right) \times h$$

- **L** = lower boundary of median class = **20**
- **f** = frequency of median class = **14**
- **CF** = cumulative frequency **before** median class = **14**
- **N/2** = **20**
- **h** = class interval = 20-10 = 10

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Step 4: Use Median Formula

Substitute the entities

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Class Interval	Frequency	Cumulative Frequency
0–10	5	5
10–20	9	14
20–30	14	28
30–40	8	36
40–50	4	40

Median Formula

$$\text{Median} = L + \left(\frac{\frac{N}{2} - CF}{f} \right) \times h$$

$$\text{Median} = 20 + \left(\frac{20 - 14}{14} \right) \times 10$$

- **L** = lower boundary of median class = **20**
- **f** = frequency of median class = **14**
- **CF** = cumulative frequency **before** median class = **14**
- **N/2** = **20**
- **h** = class interval = 20-10 = 10

Step 4: Use Median Formula

Substitute Values

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$$\text{Median} = 20 + \left(\frac{20 - 14}{14} \right) \times 10$$

$$\text{Median} = 20 + 4.29 = 24.29$$

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Key Takeaways

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- 1 Calculate cumulative frequencies
- 2 identify the class where median is located ($N/2$)
- 3 Identify the required entities from the table to put-up in formula: L, CF, f, h
- 4 Substitute the required entities in formula
- 5 Get the median value

Best thing is:

The same formula structure is applied to get any percentile values

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Quartile Formula for Grouped Data

General form

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Main Formula

$$Q_k = L + \left(\frac{\frac{kN}{4} - CF}{f} \right) \times h$$

- L = lower boundary of quartile class
- CF = cumulative frequency before quartile class
- f = frequency of quartile class
- h = class width
- $k = 1, 2, 3$

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Quartiles in Grouped Data

Understanding the idea before calculation

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Key Idea

Quartiles divide data into four equal parts after arranging it in order.

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Position Formula for Quartiles

$$Q_k = \text{value at position } \frac{kN}{4}$$

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Quartiles in Grouped Data

Understanding the idea before calculation

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Key Idea

Quartiles divide data into four equal parts after arranging it in order.

- $Q_1 \rightarrow 25\%$ of data (First quartile), $k = 1 = N/4$

Position Formula for Quartiles

$$Q_k = \text{value at position } \frac{kN}{4}$$

Quartiles in Grouped Data

Understanding the idea before calculation

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Key Idea

Quartiles divide data into four equal parts after arranging it in order.

- $Q_1 \rightarrow 25\%$ of data (First quartile), $k = 1 = N/4$
- $Q_2 \rightarrow 50\%$ of data (Median), $k = 2 = N/2$

Position Formula for Quartiles

$$Q_k = \text{value at position } \frac{kN}{4}$$

Quartiles in Grouped Data

Understanding the idea before calculation

Key Idea

Quartiles divide data into four equal parts after arranging it in order.

- $Q_1 \rightarrow 25\%$ of data (First quartile), $k = 1 = N/4$
- $Q_2 \rightarrow 50\%$ of data (Median), $k = 2 = N/2$
- $Q_3 \rightarrow 75\%$ of data (Third quartile), $k = 3 = \frac{3}{4}N$

Position Formula for Quartiles

$$Q_k = \text{value at position } \frac{kN}{4}$$

Quartiles in Grouped Data

Understanding the idea before calculation

Key Idea

Quartiles divide data into four equal parts after arranging it in order.

- $Q_1 \rightarrow 25\%$ of data (First quartile), $k = 1 = N/4$
- $Q_2 \rightarrow 50\%$ of data (Median), $k = 2 = N/2$
- $Q_3 \rightarrow 75\%$ of data (Third quartile), $k = 3 = \frac{3}{4}N$

Position Formula for Quartiles

$$Q_k = \text{value at position } \frac{kN}{4}$$

- Q_1 : use $\frac{N}{4}$

Quartiles in Grouped Data

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Position Formula for Quartiles

$$Q_k = \text{value at position } \frac{kN}{4}$$

- Q_1 : use $\frac{N}{4}$
- Q_2 : use $\frac{N}{2}$ (same as median)

Quartiles in Grouped Data

Understanding the idea before calculation

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Position Formula for Quartiles

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- Q_2 : use $\frac{N}{2}$ (same as median)
- Q_3 : use $\frac{3}{4}N$

Step 5: Use Quartile Formula

Substitute the entities for Q1

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Class Interval	Frequency	Cumulative Frequency
0–10	5	5
10–20	9	14
20–30	14	28
30–40	8	36
40–50	4	40

Quartile Formula

$$Q_1 = L + \left(\frac{\frac{N}{4} - CF}{f} \right) \times h$$

$$Q_1 = 10 + \left(\frac{10 - 5}{9} \right) \times 10$$

$$Q_1 = 10 + 5.55 = 15.55$$

- L = lower boundary of Q1 class = 10
- f = frequency of Q1 class = 9
- CF = cumulative frequency before Q1 class = 5
- $N/4 = 10$ $h = 10$

Step 6: Use Quartile Formula

Substitute the entities for Q3

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Class Interval	Frequency	Cumulative Frequency
0–10	5	5
10–20	9	14
20–30	14	28
30–40	8	36
40–50	4	40

Quartile Formula

$$Q_3 = L + \left(\frac{\frac{3N}{4} - CF}{f} \right) \times h$$

$$Q_3 = 30 + \left(\frac{30 - 28}{8} \right) \times 10$$

$$Q_3 = 30 + 2.5 = 32.5$$

- L = lower boundary of Q3 class = **30**
- f = frequency of Q3 class = **8**
- CF = cumulative frequency before Q3 class = **28**
- $3N/4 = 30$ $h = 10$

Standard Deviation

Understanding the idea first

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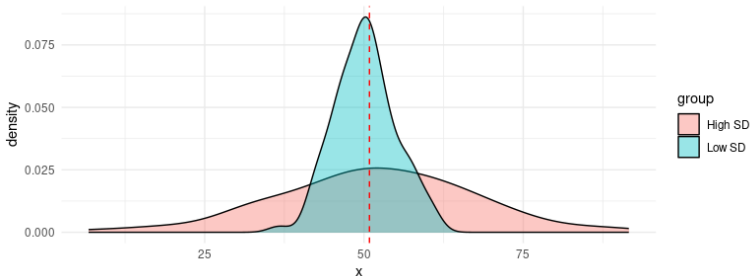
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Meaning

- Standard deviation (σ) measures how much the data values deviate from the mean.
- deviation from mean = **spread or dispersion** of the data.
- Small SD $\rightarrow$ data is closely packed around the mean
- Large SD $\rightarrow$ data is widely spread

Mean vs Spread (Standard Deviation Idea)



Standard Deviation

Using the same dataset

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Class Interval	Frequency (f)	Midpoint (x)
0–10	5	5
10–20	9	15
20–30	14	25
30–40	8	35
40–50	4	45

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Step 1: Mean

$$\bar{x} = \frac{\sum fx}{N}$$

$$\bar{x} = \frac{970}{40} = 24.25$$

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Standard Deviation

Step-by-step calculation

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Formula

$$\sigma = \sqrt{\frac{\sum f(x - \bar{x})^2}{N}}$$

- $N = 40$
- $\bar{x} = 24.25$

Step 1: Table of calculations

Class	f	x	$(x - \bar{x})$	$f(x - \bar{x})^2$
0-10	5	5	-19.25	$5(370.56)=1852.8$
10-20	9	15	-9.25	$9(85.56)=770.0$
20-30	14	25	0.75	$14(0.56)=7.84$
30-40	8	35	10.75	$8(115.56)=924.5$
40-50	4	45	20.75	$4(430.56)=1722.2$
				$\sum f(x - \bar{x})^2 = 5277.3$

Step 8: Standard Deviation

Substitute into formula

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Formula

$$\sigma = \sqrt{\frac{\sum f(x - \bar{x})^2}{N}}$$

- $N = 40$
- $\bar{x} = 24.25$

Final Result

$$\sigma = \sqrt{\frac{5277.3}{40}}$$

$$\sigma = \sqrt{131.93}$$

$$\sigma \approx \boxed{11.49}$$

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Mean

Concept and Mathematical Formulation

Intuition

Balance point of the data — if we redistribute values equally.

What we do:

- 1 Add all values
- 2 Divide by number of observations

Exam Formula

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Mapping:

- $\sum x_i$ = add everything
- n = number of observations
- Example: 2, 4, 6, 8 => $n = 4$
- $\frac{1}{4} \times \sum_1^4 2 + 4 + 6 + 8$

Median (Raw Data)

Concept and Mathematical Formulation

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Intuition

Middle value after sorting — splits data into 50% and 50%.

What we do

- Sort data in ascending order (n = total number of observations)
- When n is odd → Median = value at middle location
- When n is even → average of two middle values

Exam Formula

$$x_{0.5} = \begin{cases} x_{\frac{n+1}{2}} & \text{if } n \text{ odd} \\ \frac{x_{n/2} + x_{n/2+1}}{2} & \text{if } n \text{ even} \end{cases}$$

Mapping:

- $x_{0.5}$ = median
- Position depends on n

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Median (Grouped Data)

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Intuition

Find where cumulative frequency crosses 50%, then go inside the class.

What we do

- Identify median class:: (cum. freq ≥ 0.5)= $C_i = \sum_{j=1}^i r_j \geq 0.5$
- Interpolate within the class

Exam Formula

$$x_{0.5} = u_{i-1} + \frac{0.5 - \sum_{j=1}^{i-1} r_j}{r_i} (u_i - u_{i-1})$$

Mapping:

- u_{i-1} = lower bound
- $\sum r_j$ = cumulative frequency before the identified median class
- r_i = class frequency
- width / class interval = $(u_i - u_{i-1})$

Mode (Unclassified & Classified)

Concept and Mathematical Formulation

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Intuition

Mode = the most frequent value
“What appears the most in the data?”

Unclassified Frequency Table (Raw / Grouped Data)

$$x_M = a_i \quad \text{with} \quad r_i = \max_j(r_j)$$

Meaning:

- a_i = characteristic value
- n_i = absolute frequency (count)
- r_i = relative frequency (percentage)

Connection:

- Just pick the value with the highest count
- Largest n_i or $r_i \rightarrow$ Mode

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Mode (Unclassified & Classified)

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Classified Frequency Table (Classified Frequency Data)

$$x_M = m_j \quad \text{with} \quad \frac{r_j}{u_j - u_{j-1}} = \max_j \left(\frac{r_j}{u_j - u_{j-1}} \right)$$

Meaning:

- $[u_{j-1}, u_j)$ = class interval
- r_j = relative frequency of class
- $(u_j - u_{j-1})$ = class width
- m_j = midpoint of modal class

Connection:

- Identify class with highest **frequency density**
- Mode = midpoint of that class

Note:

- If multiple values/classes share highest frequency → no unique mode

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Quantiles and Percentiles — Concept and Formal Definition

Concept

Quantiles divide the data into proportions.

Examples:

- Median: 50% of data lies below
- Quartiles: 25%, 50%, 75%
- Percentiles: general case (any $p \in [0, 1]$)

Computation

- Use cumulative frequency (or cumulative distribution)
- Identify the smallest value where cumulative frequency reaches or exceeds p

Formal Definition

$$x_p = \min \{x : F(x) \geq p\}$$

Interpretation:

- p = proportion of data (e.g., 0.25, 0.5, 0.75)
- x_p = value below which at least $100p\%$ of observations lie

Variance and Standard Deviation

Shortcut Formula

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Formula

$$s^2 = \frac{n}{n-1} \left(\frac{\sum x_i^2}{n} - \bar{x}^2 \right)$$

$$s = \sqrt{s^2}$$

- $\sum x_i^2$ = sum of squared observations
- $\bar{x}$ = mean
- n = number of observations
- $\frac{n}{n-1}$ = correction factor

Example: Patient Dataset

Step-by-step calculation

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- $n = 10$
- $\bar{x} = 37$
- $\sum x_i^2 = 18058$

Step 1: Compute mean of squares

$$\left[\frac{18058}{10} = 1805.8 \right]$$

Step 2: Subtract square of mean

$$\left[1805.8 - 37^2 = 1805.8 - 1369 = 436.8 \right]$$

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Example: Patient Dataset

Final calculation

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Step 3: Apply correction factor

$$[s^2 = \frac{10}{9} \times 436.8 = 485.33]$$

Step 4: Standard deviation

$$[s = \sqrt{485.33} \approx 22.03]$$

Result:

- Variance = 485.33
- Standard deviation = 22.03

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Example: Oktoberfest Dataset

Step-by-step calculation

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Mapping to the
Formulas

- $n = 128$
- $\bar{x} = 3.5$
- $\sum x_i^2 = 1766$

Step 1: Compute mean of squares

$$\left[\frac{1766}{128} = 13.796875 \right]$$

Step 2: Subtract square of mean

$$\left[13.796875 - 3.5^2 = 13.796875 - 12.25 = 1.546875 \right]$$

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Example: Oktoberfest Dataset

Final calculation

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Step 3: Apply correction factor

$$[s^2 = \frac{128}{127} \times 1.546875 = 1.5591]$$

Step 4: Standard deviation

$$[s = \sqrt{1.5591} \approx 1.2486]$$

Result:

- Variance = 1.5591
- Standard deviation = 1.2486

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Variance and Standard Deviation

Two Methods

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Method 1: Table Method

$$[s^2 = \frac{\sum f(x - \bar{x})^2}{N}]$$

- Uses deviations from mean
- Conceptually clear
- Longer calculations

Method 2: Shortcut Method

$$[s^2 = \frac{n}{n-1} \left(\frac{\sum x_i^2}{n} - \bar{x}^2 \right)]$$

- Uses squares directly
- Faster computation
- Less intuitive

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